

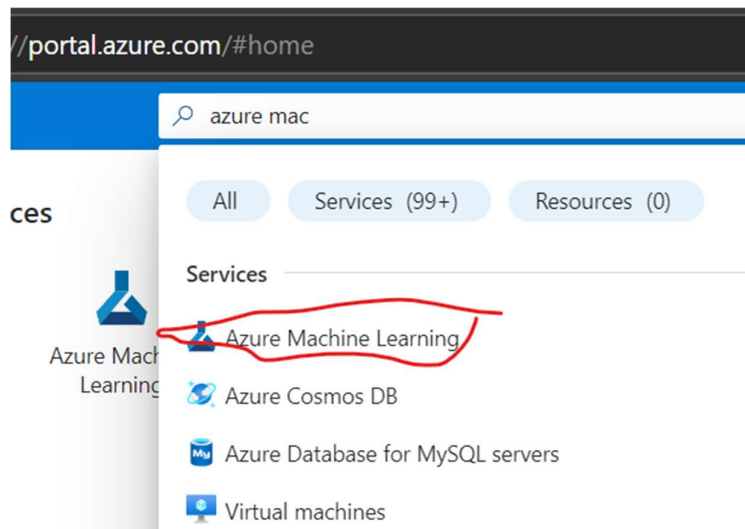
Azure Machine Learning Lab

We will take the loandata.csv provided to you and build an ML Model using ML Pipeline in Azure ML Studio, and deploy it as a web service using Inference Pipeline.

STEP 1: GOTO AZURE ML RESOURCE

Login to <https://portal.azure.com>

Search for *Azure Machine Learning*



Use the Workspace visible

STEP 2: LAUNCH AZURE ML STUDIO

Gtp Azure Machine Learning resource you created.

Click on “Launch Studio”

Machine Learning /

Machine Lea...
om (tleafservices.com)

Recently deleted ...

/ field...

...

1 of 1 >

sunita ☆ ...
Azure Machine Learning workspace

Search

Download config.json Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Events
 - Settings
 - Monitoring
 - Automation
 - Support + troubleshooting

Location
East US 2

Subscription
[Azure subscription 1](#)

Storage
[sunita4354288444](#)

Provisioning State
Succeeded

Container Registry
[f3a9951523184aa5a1b275a57b575ae1](#)

Key Vault
[sunita3509632984](#)

Application Insights
[sunita8636738174](#)

MLflow tracking URI
[azureml://eastus2.api.azureml.ms/mlflow/](#)



Work with your models in Azure Machine Learning Stud

The Azure Machine Learning Studio is a web app where you can build, train, test, and deploy ML model:
start exploring, or [learn more about the Azure Machine Learning studio](#)

[Launch studio](#)

STEP 3: INGEST YOUR DATASET

We will upload the loandata.csv provided to you

1

Azure AI | Machine Learning Studio

tleaifservices.com > sunita > Data

Data

Data assets Datastores Dataset monitors PREVIEW

Data assets are immutable references to your data that can be created from datastore deleted, but you can up-version or archive them for easy referencing and reuse in ma asset and all metadata. [Learn more about data assets](#)

+ Create Refresh Archive Reset view

Search

Name	☆	Source
loandatanew		This workspace
TD-Loan_Approval_ML_Pipeline-		This workspace
MD-Loan_Approval_ML_Pipeline		This workspace
loandata2		This workspace
sentimentdata		This workspace

Set the name and type for your data asset

▼

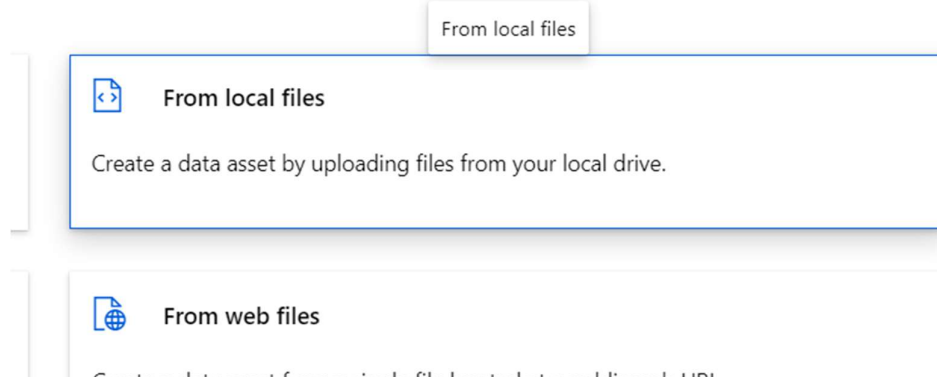
Data asset types (from Azure ML v2 APIs)

- File (uri_file)
- Folder (uri_folder)
- Table (mltable)

Dataset types (from Azure ML v1 APIs)

- Tabular
- File
- Tabular

from a local storage location on your computer, from an attached datastore, from Azure

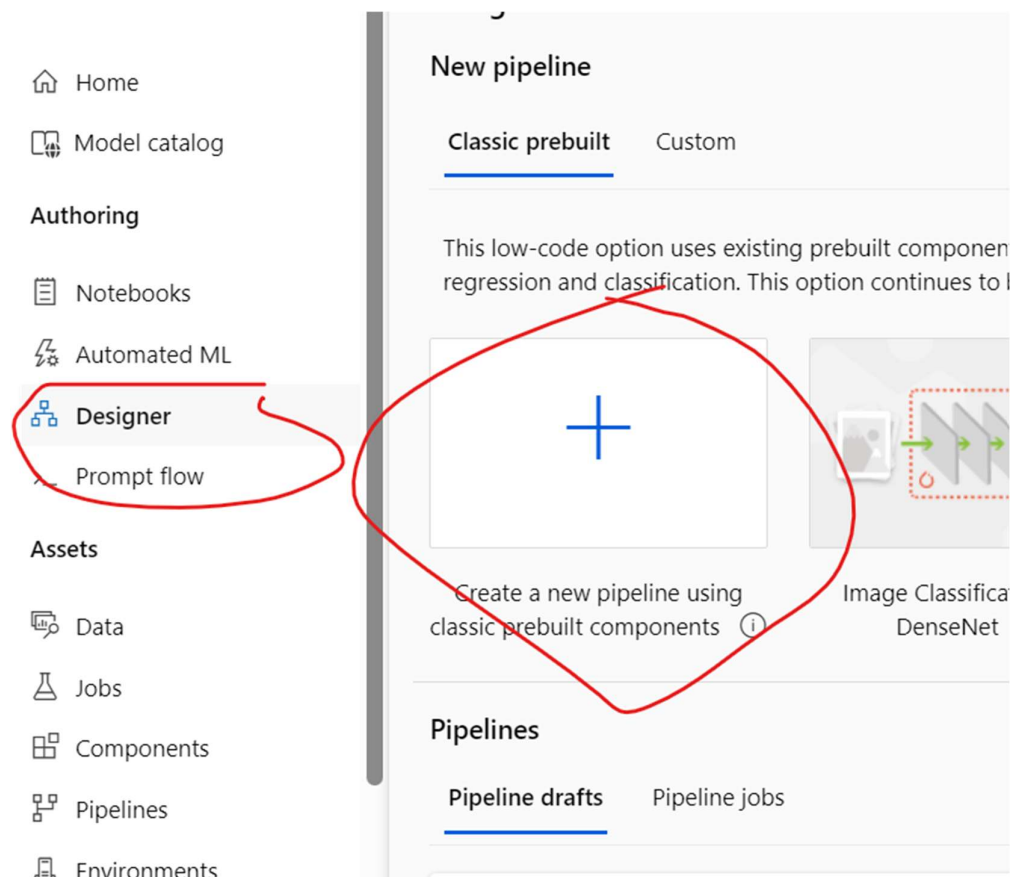


Upload the loandata csv file

The data should show up properly

STEP 4: DESIGN ML TRAINING PIPELINE

Goto Designer to design the pipeline

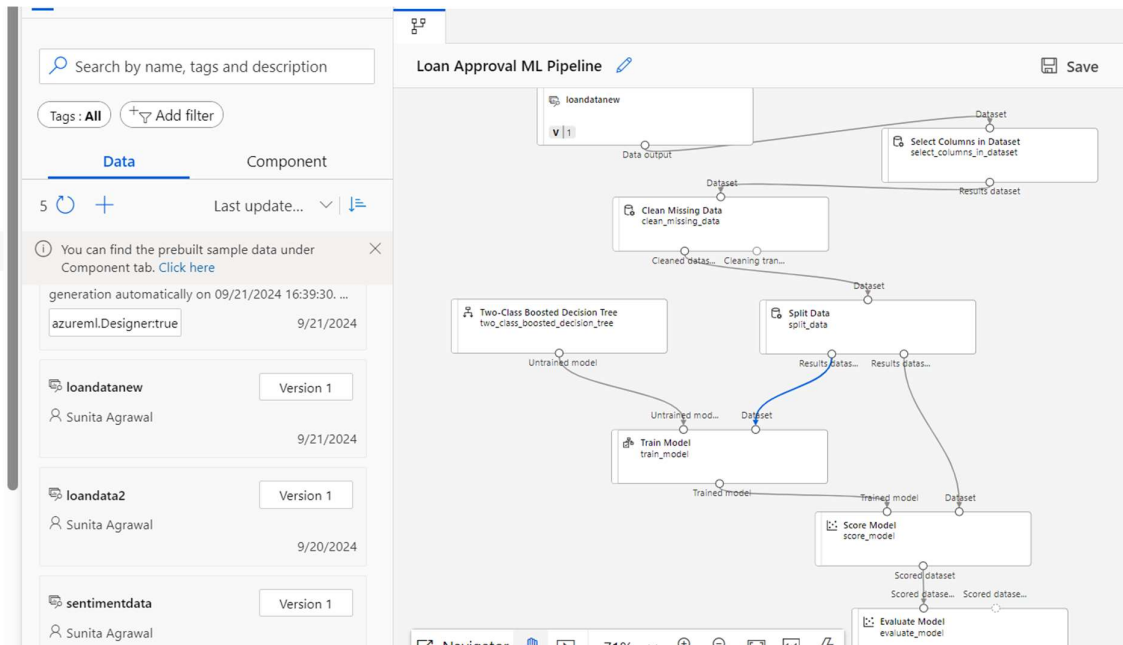


DnD the data and various components and wire them together

The pipeline should look like below

Steps:

- 1) From “Data” tab, drag and drop your dataset
- 2) From Components tab, select “Select columns from dataset”
- 3) Connect the line from the data to the “select columns.” component
- 4) Double click on the “Select columns” component. One by one choose all columns except ID
- 5) Other components shown below can be dragged and dropped and double clicked to configure
- 6) In the train model component, double click and specify target variable as Loanapproved (it is case sensitive so give it exactly like this)



Then click on “Configure and Submit”

In the config and submit wizard, choose

Create new experiment, give it a name such as Vijay_loanmodel_training

In the Compute dropdown, create a new Compute instance

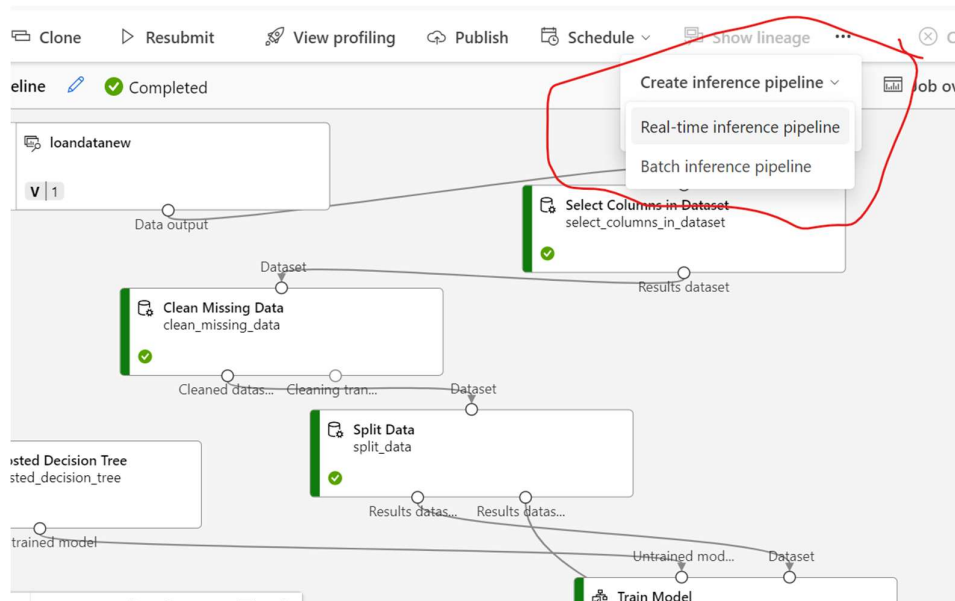
It will take some time but once the instance is ready, you can choose it and submit the job

Observe the progress of the job (You can goto the job by choosing Jobs or Pipelines section in the left pane)

Once all steps are green, you have a trained model!!

STEP 5: CREATE INFERENCE PIPELINE

From the job page where all steps are green, choose

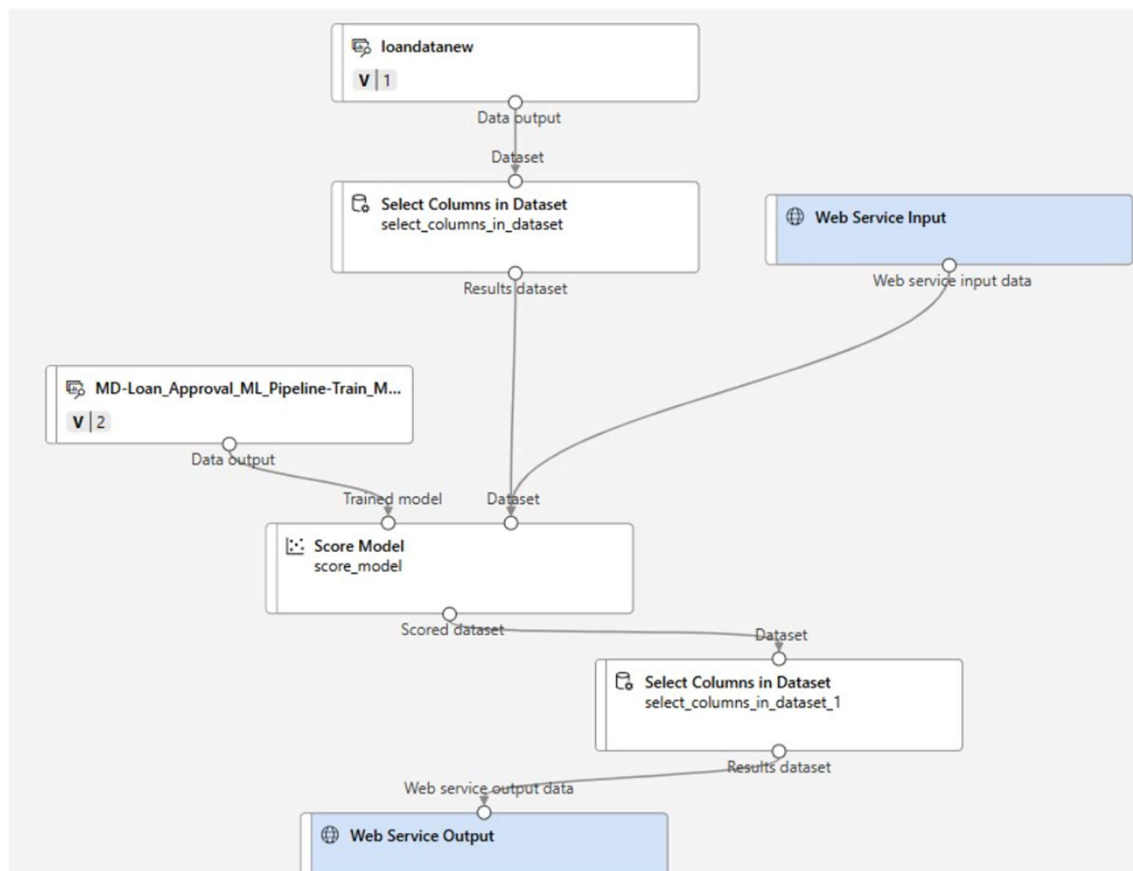


It will take you back to the Designer and already pre-populate an inference pipeline for you.

STEP 6: DESIGN THE INFERENCE PIPELINE

Inference pipeline will do the inferencing (which means when new loan application comes, it will do the prediction whether loan should be approved or not)

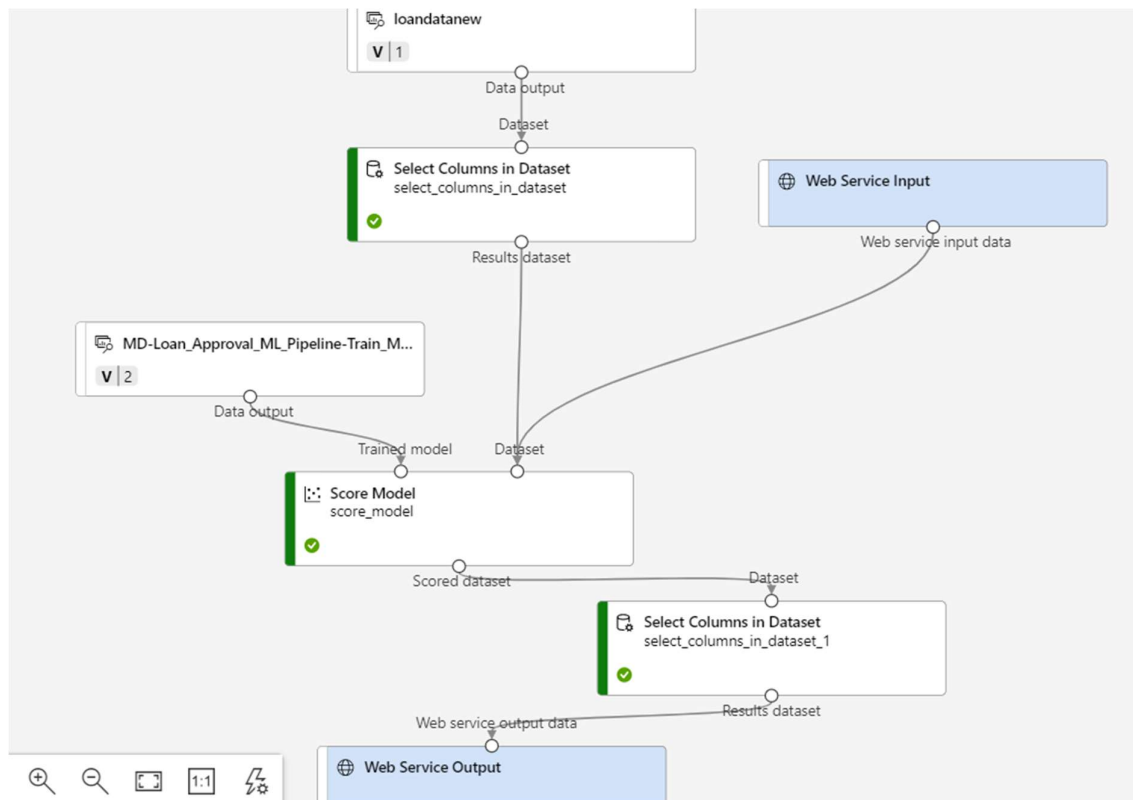
The inference pipeline should look like below:



- 1) Remove unnecessary components
- 2) Add Web Service Input
- 3) In the Select Columns in Dataset, remove the Loanapproved and ID column as those will not be provided by the new customer applying for bank loan
- 4) In the Select Columns in Dataset just before Web Service Output, choose “All Scores” This will cause only the prediction (0 or 1 in this case) to be returned from the service.

STEP 7: EXECUTE THE INFERENCE PIPELINE

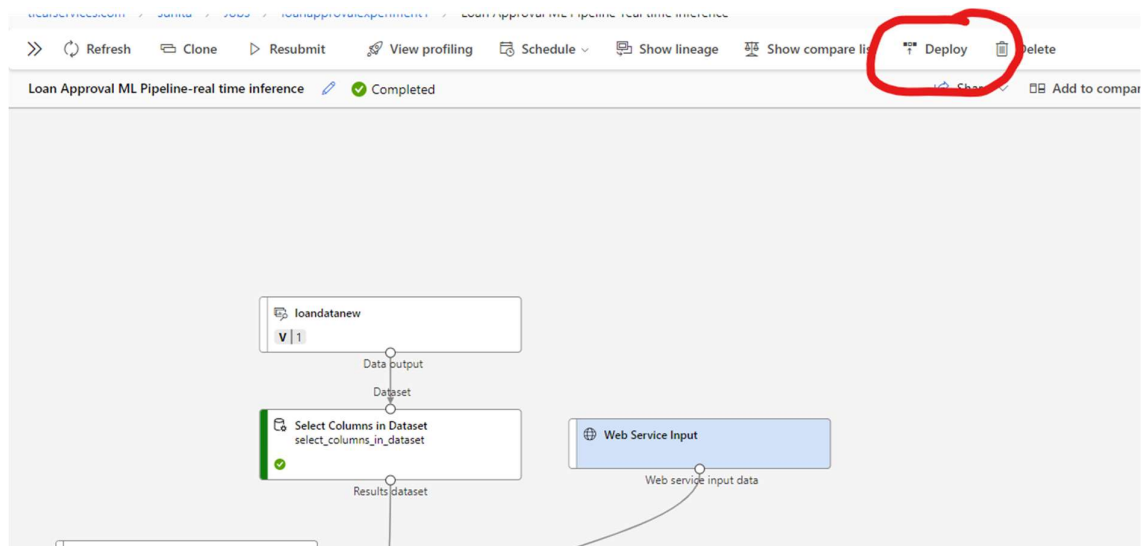
Do Configure and Submit..



STEP 8 DEPLOY THE INFERENCE PIPELINE

Once all steps are green, you need to Deploy the pipeline

This will deploy the model as a web service and create the endpoint



Set up real-time endpoint

☒ Deploy new real-time endpoint

☐ Replace an existing real-time endpoint

Name *

loanapprovalservic*e*

Description

Compute type *

Azure Container Instance

> Advanced

Deploy

Cancel

STEP 9: VALIDATE THE DEPLOYMENT

Once deployed, wait for some time

Goto “Model” in the left pane

And also “Endpoint” in the left pane

You should see the model as well as Endpoint created automatically.

The Endpoint will be in unhealthy state for some time.

Do not worry, check after some time, it will eventually become healthy.

Once healthy, note the endpoint URL and the access key:

9a: Validate the REST Endpoint:

Model catalog

Authoring

- Notebooks
- Automated ML
- Designer
- Prompt flow

Assets

- Data
- Jobs
- Components
- Pipelines
- Environments
- Models
- Endpoints**

Manage

- Compute
- Monitoring
- Data Labeling
- Linked Services

tleaifservices.com > sunita > Endpoints > loanapprovalep

loanapprovalep

Details Test Consume Logs

Compute type
AksCompute

Created by
Sunita Agrawal

Model ID
amlstudio-loanapprovalep:1

Created on
Sep 21, 2024 11:01 PM

Last updated on
Sep 21, 2024 11:20 PM

Compute target
aicluster1comput

Image ID
--

REST endpoint
<http://4.157.190.68:80/api/v1/service/loanapprovalep/score>

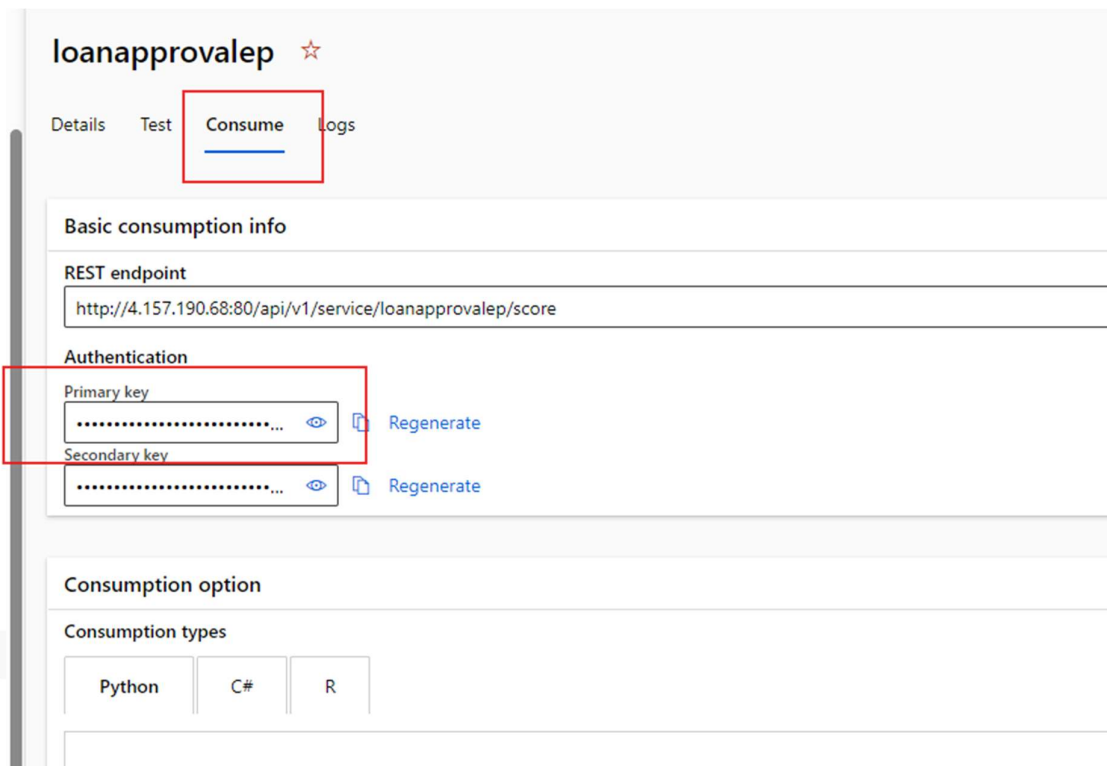
Key-based authentication enabled
true

Token-based authentication enabled
false

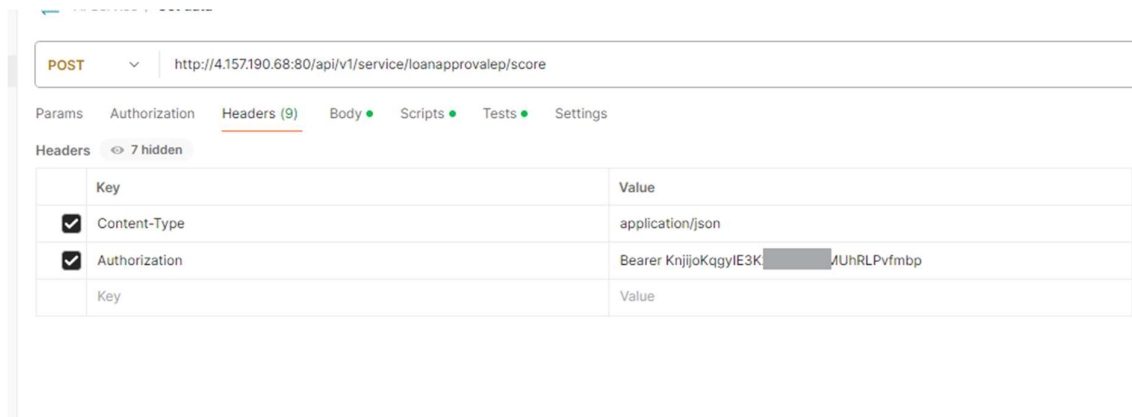
Swagger URI
<http://4.157.190.68/api/v1/service/loanapprovalep/swagger.json>

Scoring timeout

Note the Bearer token



STEP 10: Test the service using Postman

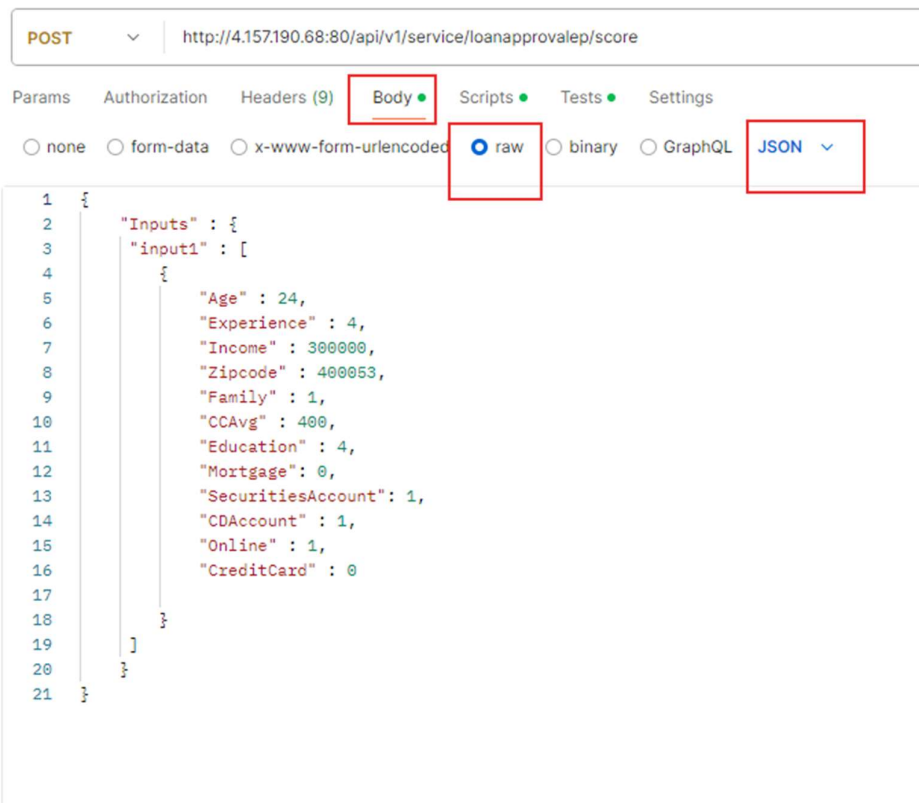


Use Post,

In Headers, put the above

Authorization needs to be:

Bearer <token>



In BODY, put following json:

```
{  
  "Inputs" : {  
    "input1" : [  
      {  
        "Age" : 24,  
        "Experience" : 4,  
        "Income" : 300000,  
        "Zipcode" : 400053,  
        "Family" : 1,  
        "CCAvg" : 400,  
        "Education" : 4,  
        "Mortgage" : 0,  
        "SecuritiesAccount": 1,  
        "CDAccount" : 1,  
        "Online" : 1,  
        "CreditCard" : 0  
      }  
    ]  
  }  
}
```

```
"CDAccount" : 1,  
"Online" : 1,  
"CreditCard" : 0  
  
}  
]  
}  
}
```

Send the request

You should see the approved or not approved prediction in the response (as 0 or 1)

STEP 11: Shutdown the service and delete the compute and other resources

Once your testing is done, shutdown the Compute and delete all resources you created including the Azure ML Workspace.

All the Best!!